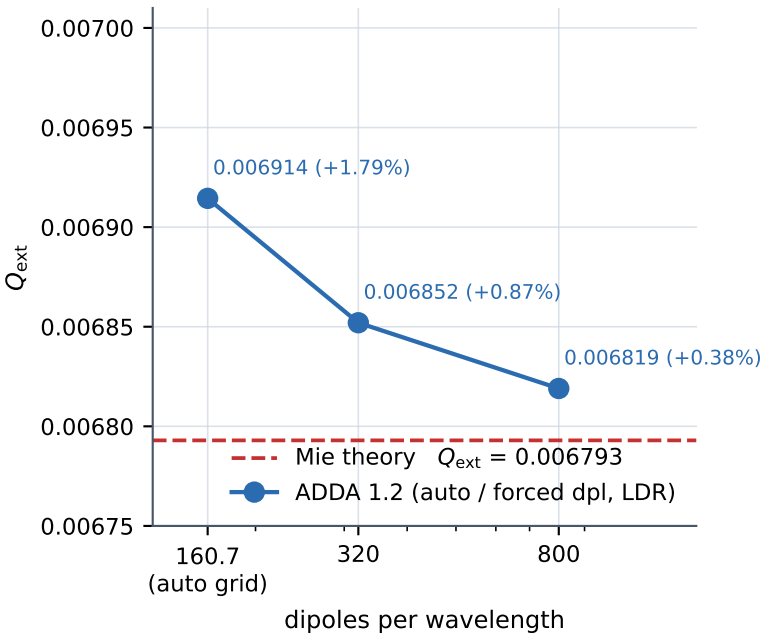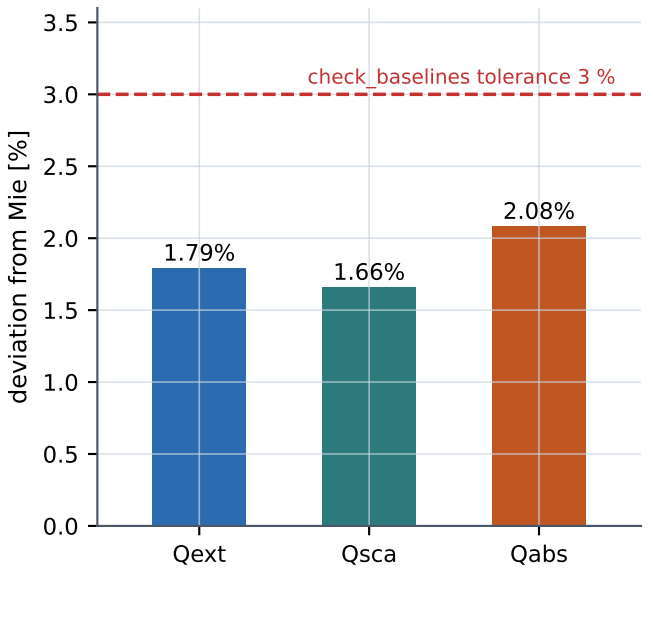Analytic validation of the sphere baseline